Notes on AI recommendations

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Codex helped with the tests, implementation and writing after Checkpoint A. This is a short decision record, not a full prompt transcript. The shared-builder suggestion came up during implementation; the other options below were evaluated while preparing these notes. These notes were also prepared with AI assistance.

Share the quote and order calculations - accepted

The starter repeats the same rules in two methods. `_build_quote` now handles both, so a pricing change only needs to be made once. It uses the state already loaded by the caller. The refactor kept the same six failing assertions before the bug fixes, and tests compare quote and order results afterward.

Use dictionary lookup everywhere - partly accepted

A dictionary is useful for adjusting stock using validated product IDs. Using it directly on incoming IDs could raise TypeError for a list or dictionary instead of the required ValueError. The small catalog scan was kept for validation. Explicit type validation could make indexing safe too; this was a simplicity choice, not a measured performance improvement.

Use floating-point round() for the discount - rejected

`round(subtotal * 0.1)` does not express half-up rounding. The integer formula was kept. A 5,005-cent test fixture needs a discount of 501 cents, so it catches the difference. The bundled prices do not reach that half-cent case on their own.

Reuse standard delivery's threshold for express - rejected

CR-01 says the $12 express fee is never waived. A separate express branch avoids accidentally applying free standard delivery to it. High-value orders and student orders are tested to make sure the fee is neither waived nor discounted.

Add a database or delivery classes - rejected

Three delivery choices do not need a new hierarchy, and the assignment already uses JSON storage. Keeping those parts small makes the actual changes easier to check. Duplicate item lines also stay separate in the result; their quantities are combined only when checking stock.

The recommendations were checked against the specification and test results. The 49 passing tests and seven caught wrong changes support these decisions, but they are not proof that the application has no remaining bugs.